

Multiple Choice (10 marks)

1. Some recent work has found an association between obesity and
 - (a) Shorter telomere strands
 - (b) Tobacco use
 - (c) Excess alcohol use
 - (d) Skin cancers
2. The DiGeorge/Shprintzen syndrome is caused by a deletion in which chromosome?
 - (a) 4
 - (b) 7
 - (c) 15
 - (d) 22
3. NEAT may be one way in which some people are able to maintain a healthy BMI; NEAT refers to
 - (a) Eating the right kinds of low calorie foods
 - (b) Daily activities like walking, gesturing, and even fidgeting
 - (c) Intense aerobic exercise practiced at least 3 times a week
 - (d) Surgical removal of fatty tissue
4. The Apo-E 2 protein is very prevalent in people who
 - (a) Die young
 - (b) Live to be 100 or older
 - (c) Have healthy immune systems
 - (d) Also have high levels of Apo-E 4
5. What is the genetic structure of rotavirus?
 - (a) Circular DNA containing virus
 - (b) RNA virus with RT
 - (c) ds RNA virus
 - (d) ds RNA virus in segments

True / False (10 marks)

Answer True or False

6. True or False: The chemistry of DNA and RNA is the most important factor for classifying viruses.
7. True or False: Monastic orders primarily follow legalism, focusing on strict adherence to laws and regulations rather than ascetic practices.
8. True or False: The RT PCR kit is essential for the diagnosis of arenavirus infection.
9. True or False: The risk of acquiring HIV is solely determined by the viral RNA level in an infected person.
10. True or False: Cerebrospinal fluid circulates between the skull and dura mater surrounding the brain.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the embryological development of the hyoid bone, including its origins and the significance of the second and third pharyngeal arches in this process.
12. Discuss the appropriateness of using an ecologic study design to investigate the hypothesis that acetaminophen use in the first trimester is linked to neural tube defects.

End of Paper